An Analysis of The American Joint Committee on Cancer 8th Edition T Staging System for Papillary Thyroid Carcinoma

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Background: The American Joint Committee on Cancer (AJCC) removed microscopic extrathyroidal extension (ETE) from the 8th edition T staging for papillary thyroid cancer (PTC) based on increasing evidence that it is not an independent prognostic factor.

Objectives: We compared the prognostic performance of AJCC 7th (pT7) and 8th (pT8) edition T stage systems, particularly in patients ≥55 years old without macroscopic ETE or distant metastases in whom T classification affects AJCC Tumor Node Metastasis (TNM) stage.

Method: A retrospective analysis of disease-free survival (DFS) in 577 patients with PTC comparing pT8 vs pT7 using the Akaike information criterion (AIC), Harrell's C-index, and Proportion of Variation Explained (PVE).

Results: Of 105 patients with AJCC7 T3 disease, 74 were down-staged. Overall, the prognostic performance of pT7 and pT8 was similar. However, in patients ≥55 years old without macroscopic ETE or distant metastases, pT8 was inferior to pT7 on the basis of higher AIC, lower C-index (0.67 vs 0.76), and lower PVE (30% vs 45%). In this subset, microscopic ETE was associated with multiple other adverse prognostic features and reduced DFS (hazard ratio, 2.8; 95% confidence interval, 1.5 to 5.2; $P = 0.002$), irrespective of tumor size.

Discussion: In our cohort, pT8 was inferior to pT7 in patients ≥55 years old without macroscopic ETE or distant metastases in whom T classification affects TNM stage. Microscopic ETE was strongly associated with other adverse prognostic factors and reduced DFS in this patient subgroup and may be an effective surrogate for disease biology in PTC, irrespective of whether it is an independent prognostic factor. **(*J Clin Endocrinol Metab* 103: 2199–2206, 2018)**

The prognosis of papillary thyroid cancer (PTC) is excellent, with disease-specific survival rates exceeding 90%; however, recurrence remains a problem in up to 30% of patients (1, 2). Accurate staging is important for prognostication and for making decisions regarding adjuvant therapy and ongoing surveillance. Each of the last three editions of the American Joint Committee on Cancer (AJCC) Tumor Node Metastasis (TNM) Staging system for well-differentiated thyroid cancer (WDTC) have involved changes to the primary tumor staging classification, reflecting uncertainty regarding the treatment of microscopic and macroscopic extrathyroidal extension (ETE) in prognostication.

ISSN Print 0021-972X ISSN Online 1945-7197
Printed in USA
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Received 23 November 2017. Accepted 11 April 2018.
First Published Online 16 April 2018

Abbreviations: AIC, Akaike information criterion; AJCC, American Joint Committee on Cancer; ATA, American Thyroid Association; CI, confidence interval; DFS, disease-free survival; ETE, extrathyroidal extension; HR, hazard ratio; pT7, prognostic performance of American Joint Committee on Cancer 7th edition; pT8, prognostic performance of American Joint Committee on Cancer 8th edition; PTC, papillary thyroid cancer; PVE, Proportion of Variation Explained; RAI, radioactive iodine; Tg, thyroglobulin; TNM, American Joint Committee on Cancer Tumor Node Metastasis; TSH, thyroid-stimulating hormone; WDTC, well-differentiated thyroid cancer.

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Multiple studies have validated the use of macroscopic ETE as a predictor of survival (3–6). On the other hand, the notion that microscopic ETE carries a worse prognosis remains controversial. The American Thyroid Association (ATA) guidelines for predicting recurrence considers patients with microscopic ETE to be at intermediate risk of recurrence and patients with macroscopic ETE to be at high risk (7). Although the ATA guidelines are widely used to determine the risk of recurrence, the AJCC TNM staging system remains the most widely used system for WDTC cancer staging.

The current 8th edition AJCC staging system for WDTC has removed microscopic ETE from the definition of pT3 disease. T3a and T3b categories were introduced for tumors >4 cm limited to the thyroid gland and tumors with macroscopic ETE limited to the strap muscles, respectively (8). Patients with microscopic ETE are hence redistributed from T3 in the 7th edition staging to T1-T3a based on tumor size in the 8th edition. These changes reflect evidence that microscopic ETE cannot be reliably identified on histopathology, and that, when present, its status as an independent prognostic factor in PTC has been increasingly questioned (9–13). This is a departure from the 6th and 7th editions of the AJCC staging, in which ETE was distributed across T3, T4a, and T4b categories to reflect minimal ETE (including microscopic ETE and macroscopic ETE into strap muscles); gross ETE involving subcutaneous soft tissue, the aerodigestive tract or recurrent laryngeal nerve; and gross ETE involving the prevertebral fascia, carotid artery, or mediastinal vessels, respectively (14).

Previous changes to the AJCC T staging models have not consistently translated to improved prognostic performance of the overall TNM staging system (15–17). To our knowledge, the effectiveness of the 8th Edition T staging system has not been independently assessed and validated. The primary aim of this study was to compare the prognostic performance of the AJCC 7th and 8th edition primary tumor staging categories (pT7 and pT8, respectively). Because the 8th edition staging also incorporates modifications to TNM groupings, including changing the age cutoff from 45 to 55 years, our secondary aim was to assess the subset of patients in whom changes to T stage leads to a change in TNM stage. This subgroup included patients ≥55 years of age without macroscopic ETE or distant metastases.

Patients and Methods

Study population

The study population was identified from a retrospective database review of all patients undergoing surgery for PTC from 1991 to 2016 at Liverpool Hospital, Sydney, Australia. We excluded patients if they had inadequate information to determine the T stage in AJCC 7th and 8th editions. The study was approved by the Area Health Service Ethics Review Board. Demographic, pathological, treatment, and follow-up data were collected for all patients. Macroscopic ETE was defined by local invasion appreciable intraoperatively or on gross inspection of the specimen at histopathology. Microscopic ETE was defined by tumor extension into the strap muscles or perithyroidal soft tissues identified on histopathology.

Surgical treatment and adjuvant therapy

At our institution, we generally perform total thyroidectomy for tumors measuring ≥10 mm preoperatively on ultrasound. Patients with tumors <10 mm in size underwent total or hemithyroidectomy at the discretion of the treating surgeon. During the early study period, central neck dissection was performed only when suspicious nodes were identified on preoperative imaging or by intraoperative assessment. In the later study period, prophylactic central neck dissection was generally performed in patients with tumors ≥10 mm in size at the discretion of the treating surgeon. Dissection of the lateral neck was performed only in patients with confirmed lateral neck nodal metastases based on either preoperative fine-needle aspiration biopsy or intraoperative frozen section and generally included levels II through V.

Indications for adjuvant radioactive iodine (RAI) therapy included tumors >10 mm, microscopic and macroscopic ETE, involved surgical margins, the presence of nodal metastases, and aggressive histologic variants of PTC. Thyroid-stimulating hormone (TSH) suppression was implemented for almost all patients where not contraindicated from a medical or quality-of-life perspective. We aimed for a TSH level <0.1 mIU/L in patients with high-risk disease or structurally incomplete response to treatment and a TSH level of 0.1 to 0.5 mIU/L in low- to moderate-risk disease. Patients were reviewed every 6 months for the first 2 years after surgery and every 12 months thereafter. Follow-up consisted of clinical history, examination, thyroglobulin (Tg) monitoring, and functional and structural imaging using diagnostic RAI scans and neck ultrasound. Computed tomography and positron emission tomography were used selectively if clinically indicated.

Statistical analysis

Data were collated and filtered using Microsoft Excel (Microsoft, Redmond, WA), and statistical was analysis performed using Stata version 12.0 SE (Stata Corporation, College Station, TX). All statistics were two sided, and a P value <0.05 was considered statistically significant. The clinical end point of interest was disease-free survival (DFS), calculated from the date of surgery to the date of first disease recurrence or death from any cause. Patients not experiencing this end point were censored at the time of last follow-up. Diagnosis of recurrence was defined as (1) disease confirmation on pathologic analysis of a suspicious structural lesion by cytology or excisional biopsy, (2) grossly abnormal imaging findings in the presence of rising Tg or Tg antibodies, or (3) persistent or rising Tg values that prompted additional RAI therapy (biochemical recurrence). Survival analysis was performed using univariate and multivariate Cox proportional hazards regression models. Survival curves were generated by the Kaplan-Meier method where indicated.

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The relative prognostic performance of each T staging system was evaluated using the Akaike information criterion (AIC), Harrell's Concordance Index (C-index), Proportion of Variation Explained (PVE), and visual inspection of Kaplan-Meier curves for stratification into distinct prognostic categories. The AIC expresses how well a model fits the data, with penalties for model complexity (18). The C-index provides a measure of model discrimination, with a value of 1 indicating perfect prediction and a value of 0.5 being equivalent to the toss of a coin (19). Discrimination is the ability of a model to distinguish individuals who experience the outcome from those who remain event free. For a prognostic model, the C-index is the chance that, given two individuals, one who will develop the event of interest and one who will remain event free, the prediction model will assign a higher probability of the event to the former. The relative effectiveness of a staging system compared with another can also be determined by calculating the proportion of variation in survival time explained by the model, or PVE. The PVE ranges from 0% to 100%, with a higher number indicating superiority (20).

Results

Patient demographics

The study population consisted of 577 patients, including 458 women and 119 men, with a median age of 47.7 years (range, 15 to 87 years) and mean follow-up of 4.6 years. The majority of patients underwent total thyroidectomy (93%). Central neck dissection was performed in 231 (40%) patients, generally as a prophylactic procedure, and lateral neck dissection was performed in 84 (14.6%) patients. RAI therapy was administered in 396 (68.6%) patients. In terms of histological subtype, classic PTC was diagnosed in 434 (75%) patients, follicular variant in 120 (21%) patients, oncocytic variant in 13 (2%) patients, diffuse sclerosing in 6 (1%) patients, and tall cell variant in 4 (1%) patients. There were 69 recurrences after a median of 1.6 years. This included 41 locoregional, 8 distant metastases, and 20 biochemical recurrences requiring additional RAI therapy. Relevant demographic and clinicopathological details are summarized in Table 1.

Table 1. Patient Clinicopathological Data

Variable	n	%
Age, y		
<55	411	71.4
≥55	165	29.6
Sex		
Female	458	79.4
Male	119	20.6
Subtype		
Conventional PTC	434	75.2
Follicular variant PTC	120	20.8
Oncocytic variant PTC	13	2.3
Diffuse sclerosing variant PTC	6	1
Tall cell variant PTC	4	0.7
Tumor size, cm		
≤2	439	76.5
2–4	105	18.3
>4	30	5.2
ETE		
No ETE	451	78.1
Microscopic ETE	80	13.9
Macroscopic ETE	46	8
Multifocality		
No	386	66.9
Yes	191	33.1
Distant metastasis at first RAI scan		
No	566	98.1
Yes	11	1.9
AJCC 8 pathological T stage		
pT1a	254	44
pT1b	161	27.9
pT2	92	15.9
pT3a	24	4.2
pT3b	7	1.2
pT4a	39	6.8
pT4b	0	0
Pathological N stage		
pN0	406	70.4
pN1a	87	15.1
pN1b	84	14.5
AJCC 8 TNM stage		
I	516	89.6
II	41	7.1
III	14	2.4
IVb	5	0.9

T stage distribution changes in AJCC 8th edition

There were 451 (78%) patients without ETE, 80 (14%) patients with microscopic ETE, and 46 (8%) patients with macroscopic ETE (seven of these were limited to the strap muscles). Table 2 shows the pathological T stage distribution of the study cohort according to AJCC 7th and 8th editions. Of the 105 patients with T3 disease as defined by the 7th edition, 74 patients had tumors ≤4 cm in size classified as T3 based on the presence of microscopic ETE. Of these patients, 30 (28.6%) were down-staged to T1a, 27 (25.7%) to T1b, and 17 (16.2%) to T2. After redistribution, the T3a category accounted for only 4.2% of the study population in the 8th edition. We only had seven patients in the new T3b category (macroscopic ETE into strap muscles only).

DFS prognostic performance comparison

The prognostic performance outcomes of pT7 and pT8 were similar based on measures of model performance (Table 3). However, as shown in Fig. 1B, there was minimal stratification of risk between T2 and T3a in pT8, and the increase in risk between categories was nonmonotonic, with T3b disease conferring a worse prognosis than T4a. The Kaplan-Meier curves for pT7 are shown in Fig. 1A for comparison, demonstrating a monotonic increase in risk between categories. When the

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Table 2. Pathological Tumor Stage Migration

	AJCC 8th Edition						
AJCC 7th Edition	**T1a**	**T1b**	**T2**	**T3a**	**T3b**	**T4a**	**Total**
T1a	224	0	0	0	0	0	224
T1b	0	134	0	0	0	0	134
T2	0	0	75	0	0	0	75
T3	30	27	17	24	7	0	105
T4a	0	0	0	0	0	39	39
Total	254	161	92	24	7	39	577

Abbreviation: T, tumor stage.

analysis was repeated adjusting for RAI administration, we noted improvement in model performance based on all objective measures, but the results for pT7 and pT8 remained similar.

We then restricted the analysis to patients in whom the newly defined T staging will influence the AJCC 8th edition TNM Stage. Patients <55 years of age were excluded because they are classified into sages I and II based on the presence of distant metastases. Patients with macroscopic ETE were excluded because they were classified as T3b, T4a, or T4b disease and were not the subject of this study focused on the removal of microscopic ETE from T staging. Finally, patients with distant metastases were excluded because their TNM stage is determined by age irrespective of T stage. Therefore, the subgroup analysis was performed in 146 patients aged ≥55 years with no macroscopic ETE or distant metastases. This showed that pT7 is clearly superior to pT8 based on a lower AIC, higher PVE (45% vs 30%), and higher C-index (0.76 vs 0.67) (Table 3). The Kaplan-Meier curves are shown in Fig. 2. Similar results were obtained when the analysis was repeated adjusting for RAI, with pT7 demonstrating a lower AIC, a higher PVE (47% vs 36%), and a higher C-index (0.77 vs 0.72).

Microscopic ETE in pT8 stage T1a-3a disease

We evaluated the prognostic value of microscopic ETE in patients with T1a-T3a disease according to the 8th edition AJCC staging. There were 531 cases, with 80 (15%) patients demonstrating microscopic ETE. Microscopic ETE was strongly associated with other adverse prognostic factors, including lymphovascular invasion (P = 0.001), multifocality (P = 0.002), positive margins ($P < 0.001$), nodal metastasis ($P < 0.001$), and extranodal extension ($P < 0.001$) (Table 4). Similar results were obtained when the analysis was restricted to patients with stage T1-3a disease, age >55, and with no distant metastasis (Table 5). On univariate analysis, the presence of microscopic ETE was associated with reduced DFS [hazard ratio (HR), 2.2; 95% confidence interval (CI), 1.2 to 4.1; P = 0.009]. The result remained consistent in the subset of 146 patients aged ≥55 years with no distant metastases (HR, 4.4; 95% CI, 1.6 to 12.5; P = 0.005) (Fig. 3). We also found that the prognostic importance of microscopic ETE was robust to adjustment for tumor size (≤1 cm, >1 to ≤2 cm, >2 to ≤4 cm, >4 cm) in the overall T1a-3a cohort (HR, 2.0; 95% CI, 1.1 to 3.7; P = 0.024) and in the subset of patients aged ≥55 years with no distant metastases (HR,

Table 3. Comparison of DFS According to 7th and 8th Edition T Staging

Variable	HR (95% CI)	*P* Value	PVE (%)	C-index	AIC
Overall study cohort (n = 577)					
AJCC 7th edition T stage			14.5	0.64	751.8
T1b vs T1a	1.7	0.153			
T2 vs T1a	2.1	0.071			
T3 vs T1a	3.4	0.001			
T4a vs T1a	4.1	0.001			
8th edition T stage			15.3	0.64	752.1
T1b vs T1a	1.8	0.1			
T2 vs T1a	2.6	0.007			
T3a vs T1a	2.7	0.078			
T3b vs T1a	7.6	<0.001			
T4a vs T1a	3.9	0.001			
Subset ≥55 y, no macroscopic ETE and no distant metastases (n = 146)					
AJCC 7th edition T stage			44.7	0.76	120.3
T1b vs T1a	1.9				
T2 vs T1a	9.1				
T3 vs T1a	11.0				
8th edition T stage			30.2	0.67	126.1
T1b vs T1a	1.6				
T2 vs T1a	6.9				
T3a vs T1a	5.8				

Abbreviations: T, tumor stage; C-index, Harrel's Concordance Index.

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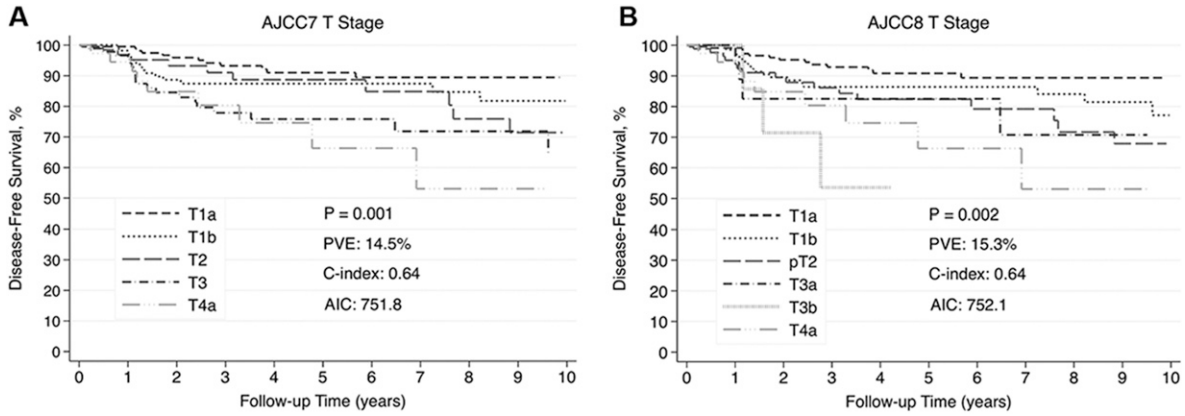

Figure 1. Kaplan-Meier curves comparing DFS when stratified by AJCC staging based on (A) the 7th edition and (B) the 8th edition.

5.3; 95% CI, 1.7 to 16.9; P = 0.004). Interaction terms and subgroup analyses by T stage similarly confirmed that the prognostic impact of microscopic ETE in these patients does not depend on tumor size.

Discussion

There is increasing evidence that microscopic ETE may not be an independent adverse prognostic factor in PTC. Based on this evidence, microscopic ETE was removed from the T category classification in the 8th edition AJCC staging for WDTC. We compared the prognostic performance of the AJCC 7th and 8th edition T staging systems in an Australian cohort of 577 patients with PTC and found that the ability of pT8 to predict DFS was similar to pT7 based on multiple objective measures of model performance. However, in the subset of patients ≥55 years old without macroscopic ETE or distant metastases with T1-3a disease, in whom changes in T stage in the 8th edition affect overall TNM stage, pT7 was clearly superior to pT8 on all performance measures, presumably due to the inclusion of microscopic ETE. These findings were robust to adjustment for RAI administration in multivariate models.

We investigated whether the finding of microscopic ETE had value in predicting recurrence. In our study population, the presence of microscopic ETE conferred a 2.24-fold increased risk of recurrence for all ages (P = 0.009) and a 2.76-fold increased risk in patients aged ≥55 with no distant metastasis (P = 0.002) when adjusted for tumor size. Although microscopic ETE was not an independent prognostic factor on a full multivariate analysis, our results support its inclusion in risk stratification models, such as the previous AJCC TNM editions and the ATA risk of recurrence guidelines (21). Microscopic ETE was strongly predictive of recurrence on univariate analysis and was an effective surrogate for other adverse risk factors, such as lymphovascular invasion, multifocality, positive margins, nodal metastasis, and extranodal extension.

Our findings on microscopic ETE are consistent with the literature. In a study of 381 patients, Park *et al.* (22) found that patients with microscopic ETE were more likely to have lymph node metastases, lymphovascular invasion, and larger tumors and carried a significantly higher rate of recurrence than patients without ETE. A multivariate analysis was not performed. In their study of 212 patients, Arora *et al.* (23) demonstrated significantly greater rates of involved surgical margins, lymph node

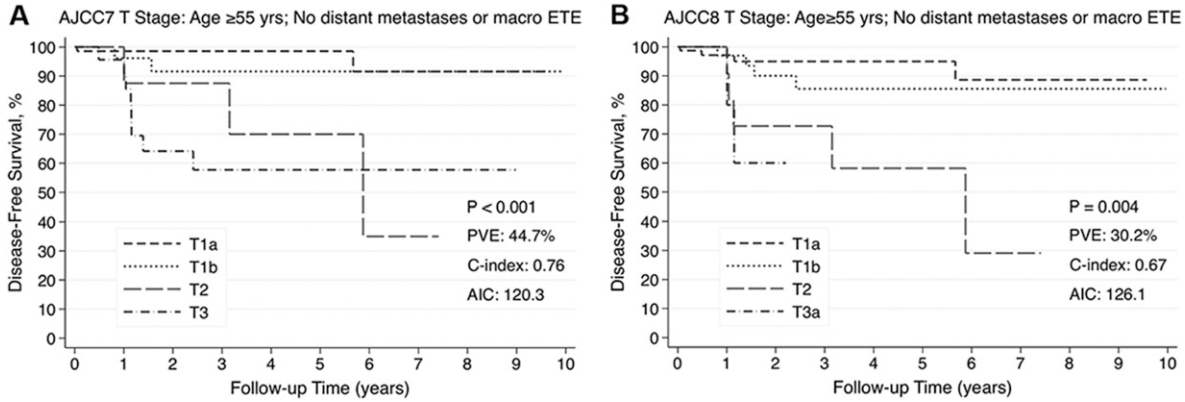

Figure 2. Kaplan-Meier curves comparing DFS in patients ≥55 years of age with no distant metastasis or macroscopic ETE when stratified by AJCC staging based on (A) the 7th edition and (B) the 8th edition.

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Table 4. Association Between Microscopic Extrathyroidal Extension and Other Risk Factors in Patients With T1-3a Disease

Variable	n (%)	No ETE, n (%)	Microscopic ETE, n (%)	*P* Value
Sex				0.104
Female	421 (79.3)	363 (80.5)	58 (72.5)	
Male	110 (20.7)	88 (19.5)	80 (27.5)	
Tumor size, cm				0.397
≤2	415 (78.2)	357 (79.2)	58 (72.5)	
2–4	92 (17.3)	75 (16.6)	17 (21.3)	
>4	24 (4.5)	19 (4.2)	5 (6.2)	
LVI				<0.001
No	488 (91.9)	423 (93.8)	65 (81.2)	
Yes	43 (8.1)	28 (6.2)	15 (18.8)	
Multifocal				0.001
No	357 (67.2)	316 (70.1)	41 (51.2)	
Yes	174 (32.8)	135 (29.9)	39 (48.8)	
Distant metastasis at first RAI				0.66
No	521 (98.1)	443 (98.2)	78 (97.5)	
Yes	10 (1.9)	8 (1.8)	2 (2.5)	
Margin status				<0.001
Clear	479 (90.2)	425 (94.2)	54 (67.5)	
Positive	52 (9.8)	26 (5.8)	26 (32.5)	
Pathological N stage				<0.001
pN0	387 (72.9)	345 (76.5)	42 (52.5)	
pN1a	74 (13.9)	64 (14.2)	10 (12.5)	
pN1b	70 (13.2)	42 (9.3)	28 (35)	
Extranodal extension				<0.001
No	515 (97)	445 (98.7)	70 (87.5)	
Yes	16 (3)	6 (1.3)	10 (12.5)	

Abbreviations: LVI, lymphovascular invasion; N, nodal.

metastases, extranodal extension, lymphovascular invasion, and recurrence in the microscopic ETE group compared with ETE-negative group. Similar to our results, microscopic ETE was not an independent predictor of mortality or recurrence in this study population on multivariate analysis.

To our knowledge, only the studies by Nixon *et al.* (11), Hay *et al.* (13), and Arora *et al.* (23) have analyzed the prognostic impact of microscopic ETE. All three of these studies rejected the hypothesis that microscopic ETE predicts survival. Our findings are in line with these studies in that microscopic ETE was not an independent predictor of DFS. Many researchers therefore endorse its removal from tumor staging systems. In contrast, we believe microscopic ETE holds value in risk stratification systems as an indicator of disease biology, particularly because it has a close association with multiple other adverse risk factors. In risk stratification systems, the presence of a single prognostic marker that is highly correlated with multiple other important adverse prognostic factors can provide powerful prognostic information while maintaining a simple parsimonious model. In this sense, a nonindependent prognostic factor may remain valuable in a staging system if it provides a good surrogate marker of disease biology due to its close association with multiple other adverse factors.

There are a number of limitations of this study. First, because of the mean follow-up of 4.6 years, a proportion of late recurrences will not be captured in our dataset. Second, the study was underpowered to use the end points of overall survival, which the TNM system is designed to predict, and disease-specific survival and local recurrence, which would be of most interest. Third, our study population did not include patients with T4b disease based on macroscopic ETE involving the carotid artery, mediastinal vessels, or prevertebral fascia. However, a small minority of patients with PTC present with T4b disease, and the analysis of staging system performance is unlikely to be affected by this. Fourth, due to the retrospective study design, patients with higher T stage and microscopic ETE may have received more aggressive treatment and surveillance, which may influence the results. However, in the absence of this potential bias due to differences in treatment, the true prognostic impact of microscopic ETE may be higher. Fifth, the presence and level of TSH suppression was not included in our analysis due to complexity and incomplete data on TSH levels over time. Finally, the presence of microscopic ETE was determined from a retrospective review of histopathological reports. The quality of pathological review and reporting has improved with time. Ideally, a blinded re-review by a single pathologist could be performed to provide uniformity. This may

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Table 5. Association Between Microscopic Extrathyroidal Extension and Other Risk Factors in Patients >55 y With T1-3a Disease and No Distant Metastasis

Variable	n (%)	No ETE (%)	Microscopic ETE (%)	*P* Value
Sex				0.449
Female	109 (74.7)	94 (75.8)	15 (68.2)	
Male	37 (25.3)	30 (24.2)	7 (31.8)	
Tumor size, cm				0.538
≤2	124 (84.9)	105 (84.7)	19 (86.4)	
2–4	16 (11)	13 (10.5)	3 (13.6)	
>4	6 (4.1)	6 (4.8)	0 (0)	
LVI				0.372
No	143 (98)	122 (98.4)	21 (95.4)	
Yes	3 (2)	2 (1.6)	1 (4.6)	
Multifocal				<0.001
No	96 (65.8)	89 (71.8)	7 (31.8)	
Yes	50 (34.2)	35 (28.2)	15 (68.2)	
Margin status				<0.001
Clear	126 (86.3)	117 (94.4)	9 (40.9)	
Positive	20 (13.7)	7 (5.6)	13 (59.1)	
Pathological N stage				<0.001
pN0	114 (78.1)	106 (85.5)	8 (36.4)	
pN1a	15 (10.3)	11 (8.9)	4 (18.2)	
pN1b	17 (11.6)	7 (5.6)	10 (45.4)	
Extranodal extension				0.164
No	144 (98.6)	123 (99.2)	21 (95.4)	
Yes	2 (1.4)	1 (0.8)	1 (4.6)	

Abbreviations: LVI, lymphovascular invasion; N, nodal.

account for our conflicting results as compared with the literature.

Conclusion

In our patient cohort, the prognostic performance of the 8th edition AJCC primary tumor staging system was not superior to the 7th edition for predicting DFS. For the subgroup of patients aged ≥55 years with no macroscopic ETE or distant metastases in whom a change in T stage would result in a change in overall TNM stage, the 8th edition performance was inferior to the 7th edition. This was due to the removal of microscopic ETE from the T staging model, which we found to be a valuable predictor of DFS survival in our cohort. Despite not being an independent prognostic factor on multivariate analyses, microscopic ETE provides valuable information on disease biology, being a surrogate marker for a number of closely associated and well-established adverse features. Further studies using overall and disease-specific survival as end points for analysis are needed to confirm our findings that the prognostic performance of the 8th edition AJCC T staging for PTC is inferior to the 7th edition due to the removal of microscopic ETE.

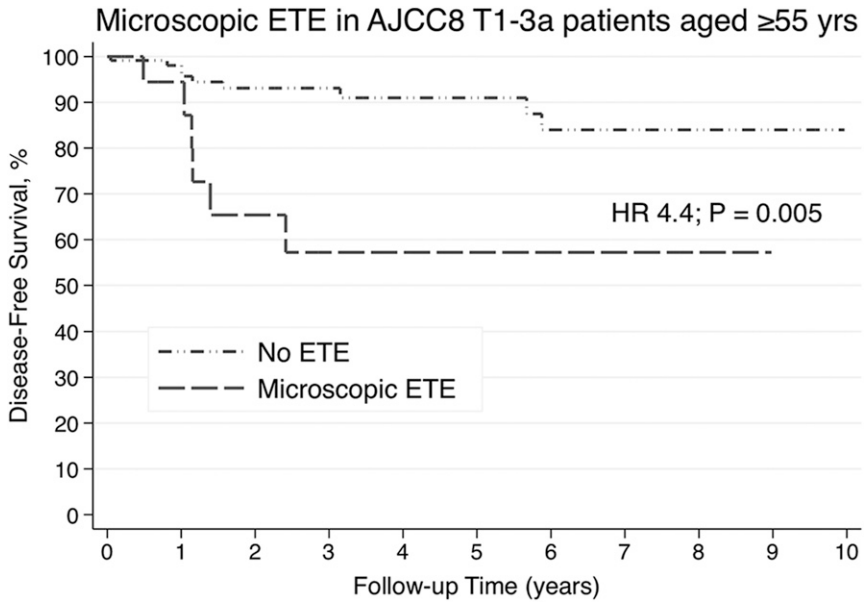

Figure 3. Kaplan-Meier curves comparing DFS in patients ≥55 years of age with and without microscopic extrathyroidal extension.

Acknowledgments

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Disclosure Summary: The authors have nothing to disclose.

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